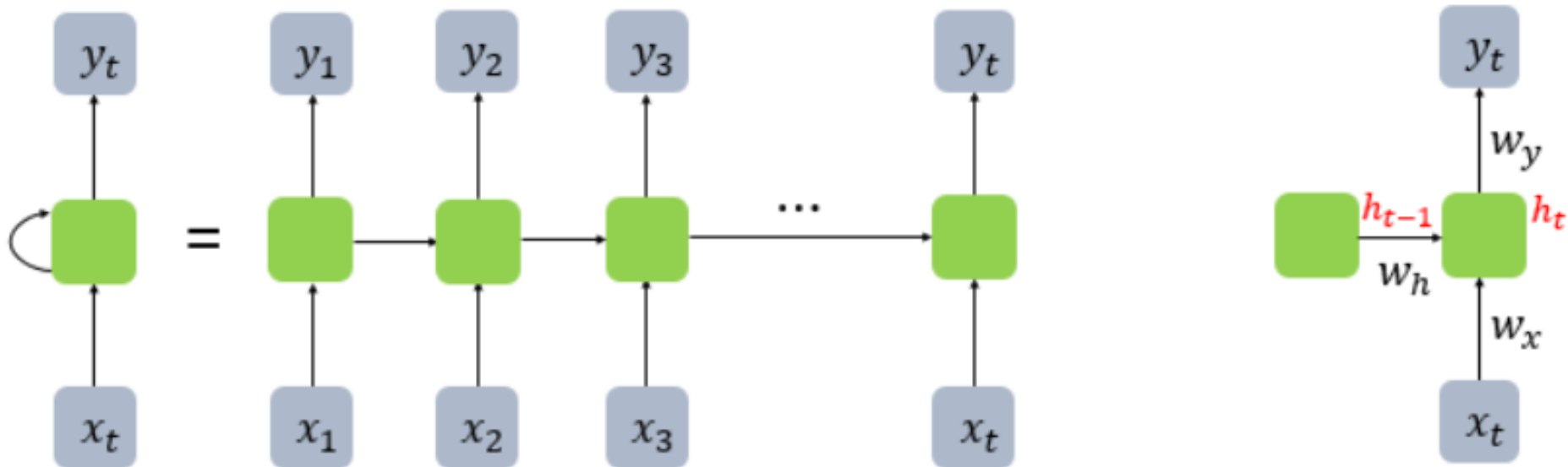


RNN (Recurrent Neural Network)

TensorFlow 2.x 강의 18] RNN 개념과 동작원리

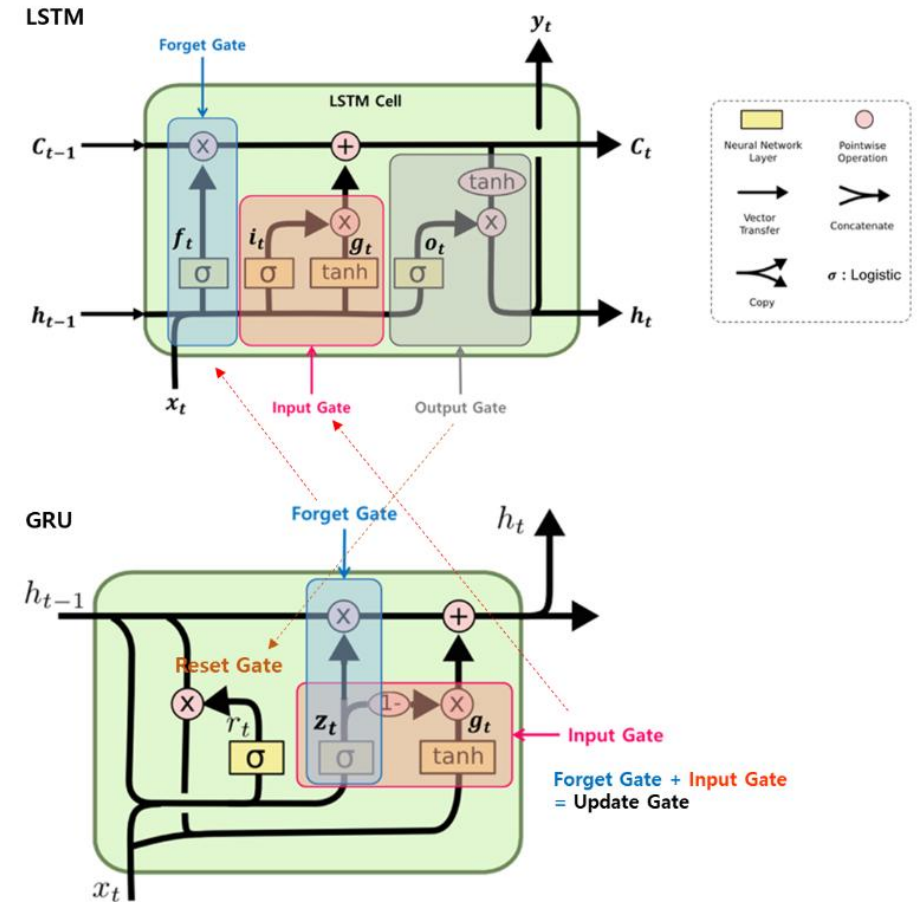
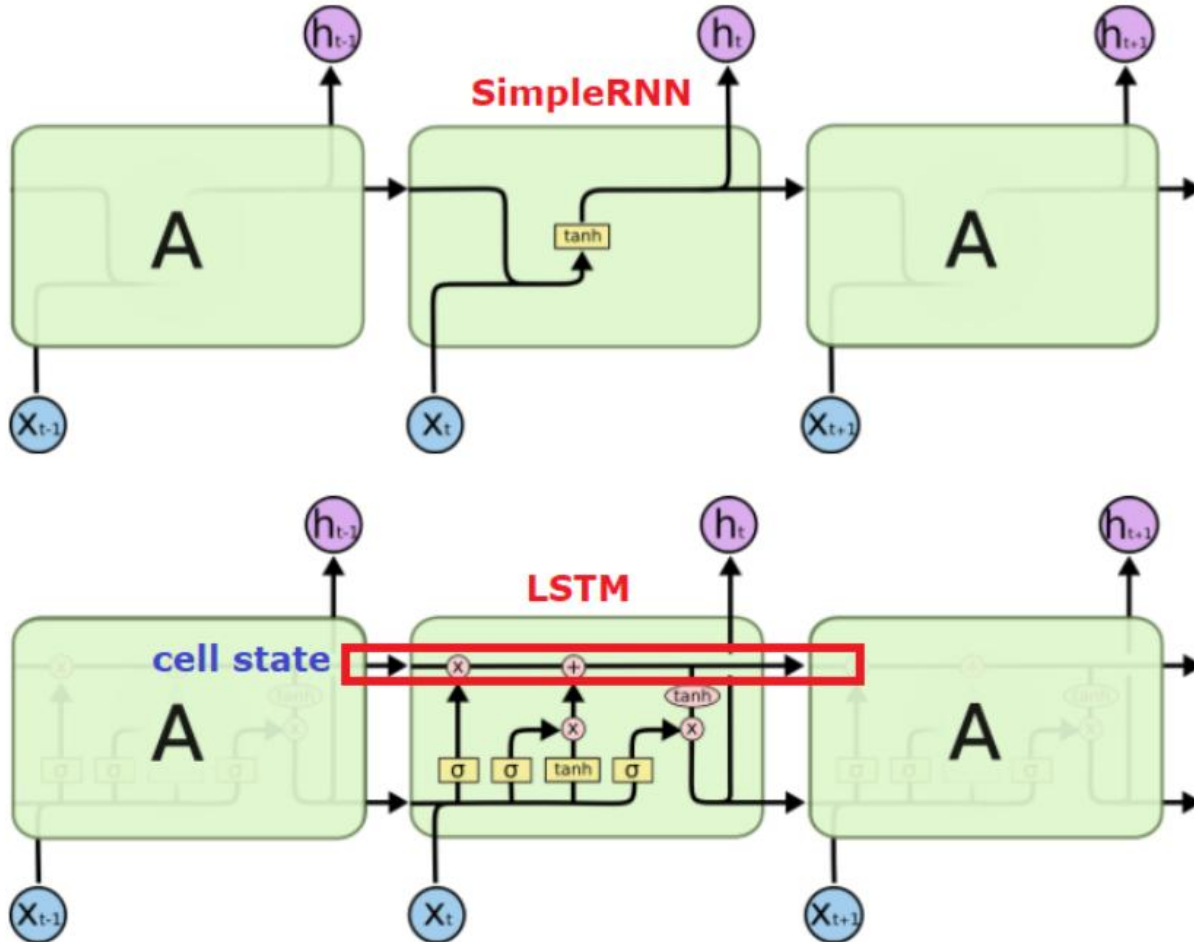
Window-size (time-steps), Horizon-factor, ...



- 은닉층 : $h_t = \tanh(W_x x_t + W_h h_{t-1} + b)$
- 출력층 : $y_t = f(W_y h_t + b)$

- Long-term dependency
- Long-distance dependency
- Long-range dependency

LSTM (Long Short-Term Memory) & GRU (Gated Recurrent Unit)



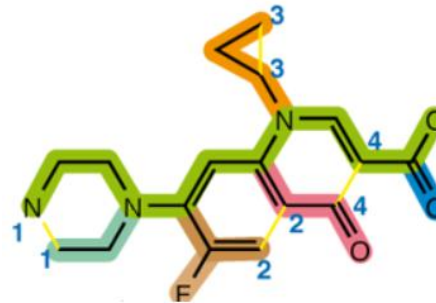
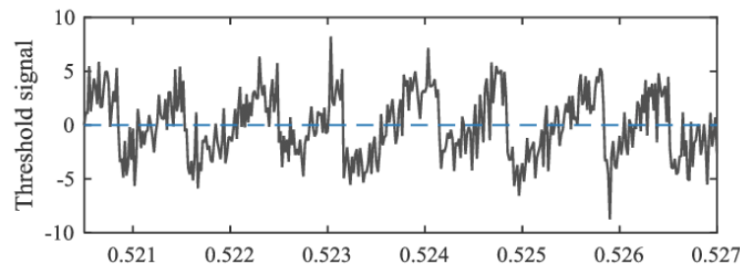
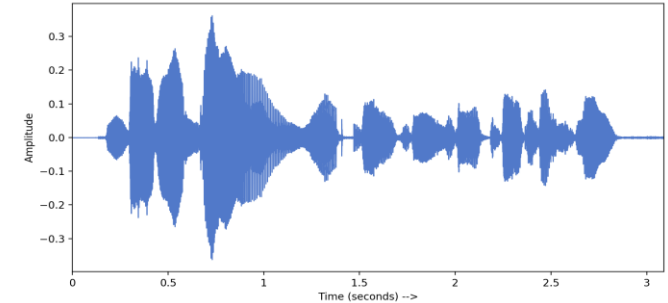
Application of RNN (Recurrent Neural Network)

Sound sequence : Voice, Music, ...

Text Sentence (→ChatGPT)

Sensor sequence signal

Molecular structure : protein folding, drug design, ...
(SMILE format, Fingerprint)



Lack of access to reading and writing put blind people at a serious

disadvantage in nineteenth-century society. Text was one of the primary methods through which people engaged with culture, communicated with others, and accessed information; without a well-developed reading system that did not rely on sight, blind people were excluded from social participation (Weygand, 2009). While

